

1. Creating and Renaming Files/Directories

Create a directory named test_dir using mkdir.

Inside test_dir, create an empty file called example.txt

Rename example.txt to renamed_example.txt using mv

```
manoj@Manojs-MacBook-Pro LINUX % cd Assignment
manoj@Manojs-MacBook-Pro Assignment % mkdir test_dir
manoj@Manojs-MacBook-Pro Assignment % ls
test_dir
manoj@Manojs-MacBook-Pro Assignment % cd test_dir
manoj@Manojs-MacBook-Pro test_dir % touch example.txt
manoj@Manojs-MacBook-Pro test_dir % ls
example.txt
manoj@Manojs-MacBook-Pro test_dir % mv example.txt renamed_example.txt
manoj@Manojs-MacBook-Pro test_dir % ls
renamed_example.txt
manoj@Manojs-MacBook-Pro test_dir %
```

2. Viewing File Contents

Use cat to display the contents of /etc/passwd

Display only the first 5 lines of /etc/passwd using head.

Display only the last 5 lines of /etc/passwd using tail.

```
manoj@Manojs-MacBook-Pro ~ % head -5 /etc/passwd
##
## User Database
##
# Note that this file is consulted directly only when the system is running
# in single-user mode. At other times this information is provided by
manoj@Manojs-MacBook-Pro ~ % tail -5 /etc/passwd
root:x:0:0:System Administrator:/var/root:/bin/sh
daemon:x:1:1:System Services:/var/root:/usr/bin/false
uucp:x:4:4:Unix to Unix Copy Protocol:/var/spool/uucp:/usr/sbin/uucico
networkd:x:24:24:Network Services:/var/networkd:/usr/bin/false
installkernel:x:25:25:Install Assistant:/var/empty:/usr/bin/false
manoj@Manojs-MacBook-Pro ~ % cat /etc/passwd
##
## User Database
##
# Note that this file is consulted directly only when the system is running
# in single-user mode. At other times this information is provided by
# Open Directory.
# See the opendirectoryd(8) man page for additional information about
# Open Directory.
##
nobody:x:-2:-2:Unprivileged User:/var/empty:/usr/bin/false
root:x:0:0:System Administrator:/var/root:/bin/sh
daemon:x:1:1:System Services:/var/root:/usr/bin/false
uucp:x:4:4:Unix to Unix Copy Protocol:/var/spool/uucp:/usr/sbin/uucico
networkd:x:24:24:Network Services:/var/networkd:/usr/bin/false
installkernel:x:25:25:Install Assistant:/var/empty:/usr/bin/false
lp:x:26:26:Printing Services:/var/spool/cups:/usr/bin/false
postfix:x:27:27:Postfix Mail Server:/var/spool/postfix:/usr/bin/false
sssd:x:31:31:Service Configuration Services:/var/empty:/usr/bin/false
cups:x:32:32:Certificate Enrollment Services:/var/empty:/usr/bin/false
appstore:x:33:33:Mac App Store Service:/var/db/appstore:/usr/bin/false
mxcl:x:54:54:IMCX AppLaunch:/var/empty:/usr/bin/false
appleevents:x:55:55:AppleEvents Daemon:/var/empty:/usr/bin/false
geod:x:56:56:Geo Services Daemon:/var/db/geod:/usr/bin/false
devdocs:x:59:59:Developer Documentation:/var/empty:/usr/bin/false
sandbox:x:60:60:Seatbelt:/var/empty:/usr/bin/false
mdnsresponder:x:65:65:mdnsResponder:/var/empty:/usr/bin/false
```

3. Searching for Patterns

Use grep to find all lines containing the word "root" in /etc/passwd

```
manoj@Manojs-MacBook-Pro ~ % grep -n root /etc/passwd
12:root:x:0:0:System Administrator:/var/root:/bin/sh
13:daemon:x:1:1:System Services:/var/root:/usr/bin/false
62:_cvmsroot:x:212:212:CVMS Root:/var/empty:/usr/bin/false
manoj@Manojs-MacBook-Pro ~ %
```

4. Zipping and Unzipping

Compress the test_dir directory into a file named test_dir.zip using zip

Unzip test_dir.zip into a new directory named unzipped_dir.

```
manoj@Manojs-MacBook-Pro Assignment % zip test_dir.zip test_dir
adding: test_dir/ (stored 0%)
manoj@Manojs-MacBook-Pro Assignment % ls
test_dir      test_dir.zip
manoj@Manojs-MacBook-Pro Assignment %
```

5. Downloading Files

Use wget to download a file from a URL (e.g., <https://example.com/sample.txt>).

```
[manoj@Manojs-MacBook-Pro Assignment % wget https://raw.githubusercontent.com/molotbot/molotbot/refs/heads/main/SECURITY.md
--2026-01-30 00:06:17-- https://raw.githubusercontent.com/molotbot/molotbot/refs/heads/main/SECURITY.md
Resolving raw.githubusercontent.com (raw.githubusercontent.com)... 2606:50c0:8000::154, 2606:50c0:8002::154, 2606:50c0:8003::154, ...
Connecting to raw.githubusercontent.com (raw.githubusercontent.com)|2606:50c0:8000::154|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 1619 (1.6K) [text/plain]
Saving to: 'SECURITY.md'

SECURITY.md                                     100%[=====]
2026-01-30 00:06:17 (5.15 MB/s) - 'SECURITY.md' saved [1619/1619]

[manoj@Manojs-MacBook-Pro Assignment % ls
SECURITY.md      test_dir        test_dir.zip
manoj@Manojs-MacBook-Pro Assignment % █
```

6. Changing Permissions

Create a file named secure.txt and change its permissions to read-only for everyone using chmod.

```
[manoj@Manojs-MacBook-Pro Assignment % touch secure.txt
[manoj@Manojs-MacBook-Pro Assignment % chmod +r secure.txt
[manoj@Manojs-MacBook-Pro Assignment % ls -la secure.txt
-rw-r--r-- 1 manoj staff 0 30 Jan 00:06 secure.txt
[manoj@Manojs-MacBook-Pro Assignment % chmod +r -w secure.txt
chmod: -w: No such file or directory
[manoj@Manojs-MacBook-Pro Assignment % chmod +r-w secure.txt
[manoj@Manojs-MacBook-Pro Assignment % ls -la
total 16
drwxr-xr-x 6 manoj staff 192 30 Jan 00:06 .
drwxr-xr-x 3 manoj staff 96 29 Jan 23:53 ..
-r--r--r-- 1 manoj staff 0 30 Jan 00:06 secure.txt
-rw-r--r-- 1 manoj staff 1619 30 Jan 00:06 SECURITY.md
drwxr-xr-x 3 manoj staff 96 29 Jan 23:54 test_dir
-rw-r--r-- 1 manoj staff 168 30 Jan 00:01 test_dir.zip
[manoj@Manojs-MacBook-Pro Assignment % ls -la secure.txt
-r--r--r-- 1 manoj staff 0 30 Jan 00:06 secure.txt
manoj@Manojs-MacBook-Pro Assignment % █
```

7. Working with Environment Variables

Use export to set a new environment variable called MY_VAR with the value "Hello, Linux!".

```
[manoj@Manojs-MacBook-Pro Assignment % export MY_VAR = 'Hello, Linux'
zsh: bad assignment
[manoj@Manojs-MacBook-Pro Assignment % export MY_VAR='Hello, Linux'
[manoj@Manojs-MacBook-Pro Assignment % echo $MY_VAR
Hello, Linux
manoj@Manojs-MacBook-Pro Assignment % █
```

Submission Guidelines -: Attach Screenshots or command along with explanation and submit in doc(google doc or microsoft doc) format also attach github repo link